

Reinforcement Learning

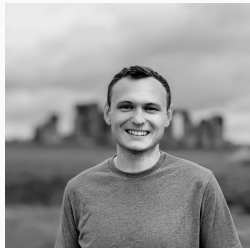
Dr Stephen James
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Imperial College London

What if machines could learn like we do?



- Humans learn through trial and error
- We learn from rewards and punishments
- We get better at tasks through practice

This course explores how we can give machines this kind of learning ability



Dr Stephen James

Background:

- 2012 — 2016: MEng in Computing at Imperial
- 2014 — 2021: Intel, Amazon, Google X, DeepMind
- 2016 — 2021: PhD in Robot Learning at Imperial
- 2021 — 2022: Postdoc at UC Berkeley
- 2022 — 2024: Director of Dyson Robot Learning Lab

Research Focus:

- Imitation learning and reinforcement learning
- Robot manipulation and whole-body control
- Vision-language-action models
- Sim-to-real transfer
- Few-shot robot learning
- 3D Vision

Gaming & Strategy (2010s)

- **2015:** DQN masters Atari games from pixels
- **2016:** AlphaGo defeats world Go champion
- **2019:** AlphaStar reaches Grandmaster in StarCraft II

Robotics (2020s)

- **2020:** Sim-to-real Quadruped locomotion
- **2022:** Robot manipulation
- **2024:** Drone racing
- **2026(?):** Humanoid whole-body control

The Foundation Model / LLM Era (2023+)

- **ChatGPT/GPT-4:** RLHF bridges gap between next-token prediction and helpful AI
- **Claude/Gemini:** Constitutional AI via RL

What do all these have in common? They learn through trial and error!



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After seeing these videos, what excites you most about RL?

- A) Gaming and competition
- B) Robotics and physical world
- C) Solving real-world problems
- D) LLMs
- E) Other
- F) All of the above!

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What's your current experience with RL?

- A) Complete beginner — never heard of it
- B) Know the basics — MDPs, Q-learning
- C) Some implementation experience
- D) Have built deep RL agents
- E) Research/industry experience

Course Timetable

Week	Schedule	Content
2	Lecture: Mon, 4-6pm Tutorial: Fri, 4-6pm	Markov Decision Processes
3	Lecture: Mon, 4-6pm Tutorial: Fri, 4-6pm	Dynamic Programming
4	Lecture: Mon, 4-6pm Tutorial: Fri, 4-6pm	Model Free Monte Carlo
5	Lecture: Mon, 4-6pm Tutorial: Fri, 4-6pm	Model Free Temporal Difference
6	Lecture: Mon, 4-6pm Tutorial: Fri, 4-6pm	Value Function Approximation Coursework Released
7	Lecture: Mon, 4-6pm Tutorial: Fri, 4-6pm	Policy Gradients
8	Lecture: Mon, 4-6pm No tutorial	Frontier
9	Lecture: Mon, 4-6pm No tutorial	Revision
10		Revision Week
11		Exam Week

⚠ Important: Let's Build This Together!

Brand New Course Last Year 🌱

This is the **second time** this version of course is running, and I'm still a relatively **new lecturer**!

My Commitment to You

- I'm working hard to make this course a success
- I'm excited to bring cutting-edge RL to you
- I'm open to trying new approaches if something isn't working

You Have the Power! 💡

- **Give me feedback early and often**; don't wait until the end
- Something unclear? Tell me immediately so I can adjust
- Have ideas for improvements? I want to hear them!

Prerequisites & Expectations

✓ What You Need

- **Programming:** Comfortable with Python
- **Mathematics:** Linear algebra, probability, calculus



Recommended Resources

- **Primary:** Sutton & Barto “Reinforcement Learning: An Introduction” (Second Edition)
- **Deep RL:** Spinning Up in Deep RL (OpenAI)
- **Papers:** Will suggest key papers throughout



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How excited are you about this course?

- A) 😄 Extremely excited, can't wait to start!
- B) 😊 Very excited, sounds interesting
- C) 😐 Moderately excited, let's see how it goes
- D) 😐 Not sure yet, need to learn more
- E) 😓 Worried, seems challenging